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MEDIUM TRANSPORT DEVICE AND IMAGE READING APPARATUS

Abstract

An ultrasonic detection unit includes a transmission sensor chip, a reception sensor chip, a transmission substrate provided with the transmission sensor chip, and a reception substrate provided with the reception sensor chip. The reception substrate is provided with an amplifier that amplifies a reception signal of an ultrasonic wave received by the reception sensor chip, and a shielding member that covers at least a part of the amplifier. A height of the reception sensor chip with respect to the reception substrate is lower than a height of the shielding member with respect to the reception substrate.

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Background/Summary

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-029619, filed Feb. 29, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a medium transport device that transports a medium and an image apparatus including the medium transport device.

2. Related Art

[0003] Hitherto, as disclosed in JP-A-2020-25242, there has been used a technique of detecting double feeding of media by using an ultrasonic sensor.

[0004] Adoption of an ultrasonic sensor easily causes an increase of a device size.

[0005] In the related art, there is room for improvement in view of device size reduction by devising a configuration of an ultrasonic sensor and arrangement thereof.

SUMMARY

[0006] In order to solve the above-mentioned problem, a medium transport device according to the present disclosure includes a first roller pair configured to transport a medium in a transport direction, a second roller pair arranged downstream of the first roller pair in the transport direction, and an ultrasonic detection unit arranged between the first roller pair and the second roller pair in the transport direction, wherein the ultrasonic detection unit includes a transmission sensor chip configured to emit an ultrasonic wave along a first axis toward a first surface of the medium being transported, a reception sensor chip being configured to receive an ultrasonic wave and being arranged at a position on the first axis so that the medium is sandwiched between the reception sensor chip and the transmission sensor chip, a transmission substrate being provided with the transmission sensor chip and including a substrate surface on which the transmission sensor chip is placed, a reception substrate being provided with the reception sensor chip and including a first substrate surface on which the reception sensor chip is placed, an amplifier being provided at the reception substrate and being configured to amplify a reception signal of an ultrasonic wave received by the reception sensor chip, and a shielding member being provided at the reception substrate and being configured to cover at least a part of the amplifier, and a height of the reception sensor chip with respect to the reception substrate is lower than a height of the shielding member with respect to the reception substrate.

[0007] Further, an image reading apparatus according to the present disclosure includes the medium transport device, and a reading unit being located downstream of the second roller pair in the transport direction and being configured to read an image of the medium.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a perspective view of a scanner as viewed from the front.

[0009] FIG. 2 is a view of a document transport path of the scanner as viewed in a width direction.

[0010] FIG. 3 is a block diagram illustrating a control system of the scanner.

[0011] FIG. 4 is a perspective view of a state in which a second unit is opened with respect to a first unit.

[0012] FIG. 5 is a plan view of the first unit.

[0013] FIG. **6** is a perspective view of a state in which a cover provided to the first unit is attached.

[0014] FIG. **7** is a perspective view of a state in which the cover provided to the first unit is removed.

[0015] FIG. **8** is a perspective view of the first unit as viewed from the back side.

[0016] FIG. **9** is a perspective view illustrating a fixed state of a transmission substrate holder.

[0017] FIG. **10** is a plan view of the second unit.

[0018] FIG. **11** is a perspective view illustrating a fixed state of a reception substrate holder.

[0019] FIG. **12** is a perspective view illustrating a positional relationship between the first roller pair, the second roller pair, the third roller pair, and the ultrasonic detection unit.

[0020] FIG. **13** is a perspective view of the ultrasonic detection unit.

[0021] FIG. **14** is a perspective view of the ultrasonic detection unit.

[0022] FIG. **15** is a perspective view of a transmission substrate.

[0023] FIG. **16** is a side view of the transmission substrate.

[0024] FIG. **17** is a perspective view of the transmission substrate holder.

[0025] FIG. **18** is a perspective view of the transmission substrate holder.

[0026] FIG. **19** is a perspective view of a protection member.

[0027] FIG. **20** is an enlarged plan view of the protection member.

[0028] FIG. **21** is a cross-sectional view illustrating a structure of a transmission sensor chip.

[0029] FIG. **22** is a cross-sectional view illustrating a configuration of an ultrasonic detection unit.

[0030] FIG. **23** is a view illustrating a positional relationship between the ultrasonic detection unit and a first roller pair.

[0031] FIG. **24** is a view illustrating a positional relationship between the ultrasonic detection unit, and a second roller pair or a third roller pair.

[0032] FIG. **25** is a view illustrating a configuration of the second roller pair.

[0033] FIG. **26** is a cross-sectional perspective view illustrating a structure of a first reading unit.

[0034] FIG. **27** is a view illustrating a positional relationship between a sensor substrate of the first reading unit, and the transmission sensor chip and the transmission substrate.

[0035] FIG. **28** is a view illustrating a second reading unit, and a reception sensor chip and a reception substrate.

[0036] FIG. **29** is a view illustrating occupied regions of the first roller pair, the second roller pair, a transport motor, the transmission sensor chip, the transmission substrate, the reception sensor chip, and the reception substrate.

[0037] FIG. **30** is a side view of the reception substrate.

[0038] FIG. **31** is a side view of the reception substrate.

[0039] FIG. **32** is a side view of the reception substrate.

DESCRIPTION OF EMBODIMENTS

[0040] Hereinafter, the present disclosure is described in brief.

[0041] A medium transport device according to a first aspect includes a first roller pair configured to transport a medium in a transport direction, a second roller pair arranged downstream of the first roller pair in the transport direction, and an ultrasonic detection unit arranged between the first roller pair and the second roller pair in the transport direction, wherein the ultrasonic detection unit includes a transmission sensor chip configured to emit an ultrasonic wave along a first axis toward a first surface of the medium being transported, a reception sensor chip being configured to receive an ultrasonic wave and being arranged at a position on the first axis so that the medium is sandwiched between the reception sensor chip and the transmission sensor chip, a transmission substrate being provided with the transmission sensor chip and including a substrate surface on which the transmission sensor chip is placed, a reception substrate being provided with the reception sensor chip and including a first substrate surface on which the reception sensor chip is placed, an amplifier being provided at the reception substrate and being configured to amplify a reception signal of an ultrasonic wave received by the reception sensor chip, and a shielding

member being provided at the reception substrate and being configured to cover at least a part of the amplifier, and a height of the reception sensor chip with respect to the reception substrate is lower than a height of the shielding member with respect to the reception substrate.

[0042] According to the present aspect, a height of the reception sensor chip with respect to the reception substrate is lower than a height of the shielding member with respect to the reception substrate. Thus, in the configuration in which the ultrasonic detection unit includes the reception substrate and the shielding member, a device dimension can be reduced.

[0043] The second aspect is an aspect dependent on the first aspect, wherein the first axis is inclined with respect to a surface of the medium passing through between the transmission sensor chip and the reception sensor chip, a thickness of the transmission sensor chip is less than a thickness of the transmission substrate, a thickness of the reception sensor chip is less than a thickness of the reception substrate, and at least a part of the transmission sensor chip and at least a part of the reception sensor chip are present within a range of the second roller pair in a normal line direction with respect to a surface of the medium at a nipping position of the second roller pair.

[0044] According to the present aspect, at least a part of the transmission sensor chip and at least a part of the reception sensor chip are present within the range of the second roller pair in the normal line direction with respect to the surface of the medium at the nipping position of the second roller pair. Thus, a device dimension in the normal line direction with respect to the surface of the medium at the nipping position of the second roller pair can be reduced.

[0045] The third aspect is an aspect dependent on the first aspect, wherein the amplifier and the shielding member are provided to a surface of the reception substrate, the surface being opposite to the surface at which the reception sensor chip is provided.

[0046] According to the present aspect, the amplifier and the shielding member are provided to a surface of the reception substrate, the surface being opposite to the surface at which the reception sensor chip is provided. Thus, the one surface and the other surface of the reception substrate can be utilized efficiently, and the reception substrate is easily reduced in size. With this, a device dimension can be reduced.

[0047] This aspect is not limited to the first aspect, and may be subordinate to the second aspect.

[0048] The fourth aspect is an aspect dependent on the first aspect, wherein the reception substrate is provided with a connector at a surface opposite to a surface facing the transmission sensor chip, and the amplifier and the shielding member are arranged at a surface of the reception substrate, the surface being provided with the connector.

[0049] According to the present aspect, the amplifier and the shielding member are arranged in a space generated by arranging the connector. Thus, a device dimension can be reduced.

[0050] This aspect is not limited to the first aspect, and may be subordinate to the second aspect.

[0051] The fifth aspect is an aspect dependent on the first aspect, wherein the amplifier is arranged at the surface of the reception substrate, the surface being a surface provided with the reception sensor chip, and the shielding member covers the reception sensor chip in addition to the amplifier.

[0052] According to the present aspect, the shielding member covers the reception sensor chip in addition to the amplifier. Thus, the same shielding member covers the amplifier and the reception sensor chip, and cost reduction can be achieved while reducing a device dimension.

[0053] This aspect is not limited to the first aspect, and may be subordinate to the second aspect.

[0054] The sixth aspect is an aspect dependent on the first aspect, wherein at least a part of the reception substrate and at least a part of the shielding member are present within any one of a range of the first roller pair in a normal line direction with respect to a surface of the medium at a nipping position of the first roller pair, and a range of the second roller pair in a normal line direction with respect to a surface of the medium at a nipping position of the second roller pair.

[0055] According to the present aspect, at least a part of the reception substrate and at least a part of the shielding member are present within any one of a range of the first roller pair in a normal line direction with respect to a surface of the medium at a nipping position of the first roller pair, and a

range of the second roller pair in a normal line direction with respect to a surface of the medium at a nipping position of the second roller pair. Thus, a device dimension in the normal line direction with respect to the surface of the medium at the nipping position of the first roller pair or the second roller pair can be reduced.

[0056] This aspect is not limited to the first aspect, and may be subordinate to any of the second to fifth aspects.

[0057] The seventh aspect is an aspect dependent on the first aspect, wherein the first axis is inclined with respect to a surface of the medium passing through between the transmission sensor chip and the reception sensor chip, and, when the ultrasonic detection unit is calibrated, the reception sensor chip receives an ultrasonic wave while a bias voltage is applied to the reception sensor chip, and the amplifier amplifies the reception signal.

[0058] According to the present aspect, when a piezoelectric element is used in the reception sensor chip, variations in the polarization direction of the piezoelectric element are less likely to occur, an efficient amplitude can be obtained, and an appropriate reception signal can be output.

[0059] This aspect is not limited to the first aspect, and may be subordinate to any of the second to sixth aspects.

[0060] The eighth aspect is an aspect dependent on the seventh aspect, wherein the first axis is inclined with respect to a surface of the medium passing through between the transmission sensor chip and the reception sensor chip, and, when the ultrasonic detection unit detects double feeding of media, the reception sensor chip receives an ultrasonic wave while a bias voltage is applied to the reception sensor chip, and the amplifier amplifies the reception signal.

[0061] According to the present aspect, when a piezoelectric element is used in the reception sensor chip, variations in the polarization direction of the piezoelectric element are less likely to occur, an efficient amplitude can be obtained, and an appropriate reception signal can be output.

[0062] An image reading apparatus according to a ninth aspect includes the medium transport device according to any one of the first to eighth aspects, and a reading unit being located downstream of the second roller pair in the transport direction and being configured to read an image of the medium.

[0063] According to the present aspect, the operational effects of any one of the first to eighth aspects described above can be obtained in the image reading apparatus.

[0064] Hereinafter, the present disclosure is described in detail.

[0065] Hereinafter, as an example of the image reading apparatus, a scanner **1** capable of reading at least one surface of a first surface **S1** and a second surface **S2** opposite to the first surface **S1** of a document **P** being an example of a medium is described. The scanner **1** is a so-called sheet-feed type scanner that reads the document **P** while moving the document **P** with respect to a first reading unit **32** and a second reading unit **33**, which are described later.

[0066] In this specification, the document **P** includes not only a sheet-like document but also a card-like document and a booklet-like document.

[0067] Note that the scanner **1** can be considered as a medium transport device **100** from the perspective of transporting the document **P** being an example of the medium. In this case, the scanner **1** is configured to include the medium transport device **100**, and the first reading unit **32** and the second reading unit **33**, which are described later.

[0068] In an XYZ coordinate system illustrated in each of drawings, an X-axis direction is a width direction of a device and a document width direction. A Y-axis direction is a depth direction of the device, and a Z-axis direction is a direction along a vertical direction. In the present exemplary embodiment, a +Y direction is a direction from a rear surface to a front surface of the apparatus, and a -Y direction is a direction from the front surface to the rear surface of the apparatus. Also, a left direction as seen from the front surface of the apparatus is a +X direction, and a right direction is a -X direction.

[0069] Also, hereinafter, a direction in which the document **P** is transported may be referred to as

“downstream”, and a direction opposite thereto may be referred to as “upstream”.

Overview of Scanner

[0070] In FIG. 1, the scanner 1 includes a device main body 2 and a main body support unit 6 that supports the device main body 2. The main body support unit 6 is placed on a placement surface GS of the device. As an example, the placement surface GS is a surface parallel to a horizontal plane.

[0071] The device main body 2 includes a first unit 3, a second unit 4, and a third unit 5.

[0072] The second unit 4 and the third unit 5 are provided to the first unit 3 rotatably about a rotary shaft (omitted in illustration) parallel to the X-axis direction. The second unit 4 and the third unit 5 can rotate integrally about the rotary shaft with respect to the first unit 3. The reference symbol 8a in FIG. 1 is an unlocking unit. A user can unlock the second unit 4 and the third unit 5 with respect to the first unit 3 by sliding the unlocking unit 8a in the -X direction. Further, when the second unit 4 and the third unit 5 rotate with respect to the first unit 3, a part of a document transport path can be exposed (see FIG. 4). In particular, when the second unit 4 is opened with respect to the first unit 3, an upstream feeding path R0, a downstream feeding path R1, and a reading transport path R2, which are described later, can be exposed.

[0073] Further, the third unit 5 can rotate with respect to the first unit 3 and the second unit 4 about a rotary shaft (omitted in illustration) parallel to the X-axis direction. When the third unit 5 rotates with respect to the second unit 4, a U-turn discharge path R3 (see FIG. 2) located downstream of the reading transport path R2, which is described later, can be exposed. In other words, the U-turn discharge path R3 is formed between the third unit 5 and the second unit 4.

[0074] Note that the third unit 5 is held with respect to the second unit 4 by a snap-fit structure (omitted in illustration). When a user applies an external force to the third unit 5, holding of the third unit 5 with respect to the second unit 4 can be canceled to open the third unit 5.

[0075] The device main body 2 is rotatable about a main body rotary shaft 6c (see FIG. 2) with respect to the main body support unit 6, and in the embodiment, the device main body 2 can hold two postures by rotating. The posture of the device main body 2 illustrated in FIG. 2 is one of the two postures, and is a normal reading posture. Note that the device main body 2 may rotate from the normal reading posture so that the reading transport path R2 approaches the horizontal plane, and may be in a booklet reading posture (omitted in illustration).

[0076] In FIG. 1, an operation unit 7 configured by a plurality of operation buttons is provided at the front surface of the device. The plurality of operation buttons are configured by operation buttons 7a, 7b, and 7c in the embodiment, and a user operation is received with those buttons.

Document Transport Path in Scanner

[0077] Next, with reference to FIG. 2, a document transport path in the scanner 1 is described. The bold broken line in FIG. 2 indicates a transport path in which the document P is transported.

[0078] The reference symbol R0 is a transport path upstream of a first roller pair 13, and is referred to as the upstream feeding path R0 hereinafter.

[0079] The reference symbol R1 is a transport path between the first roller pair 13 and a second roller pair 16, and is referred to as the downstream feeding path R1.

[0080] The reference symbol R2 is a transport path between the second roller pair 16 and a third roller pair 20, and is referred to as the reading transport path R2. Note that the reading transport path R2 is a transport path that faces the first reading unit 32 and the second reading unit 33, which are described later.

[0081] The first unit 3 forms the lower sides of the upstream feeding path R0, the downstream feeding path R1, and the reading transport path R2, and the second unit 4 forms the upper sides of the upstream feeding path R0, the downstream feeding path R1, and the reading transport path R2.

[0082] Further, the reference symbol R3 is a transport path that is inverted upward from the third roller pair 20, and is referred to as the U-turn discharge path R3. The U-turn discharge path R3 is formed between the second unit 4 and the third unit 5.

[0083] The normal reading posture (FIG. 2) of the device main body **2** is a posture in which the reading transport path **R2** is coupled to the U-turn discharge path **R3** by a flap **35**. Note that, in the booklet reading posture (omitted in illustration) of the device main body **2**, the flap **35** is in a posture indicated by the two-dot chain line, the reading transport path **R2** is not coupled to the U-turn discharge path **R3**, and the document **P** is discharged in an obliquely downward direction containing a +Y-direction component and a -Z-direction component from the reading transport path **R2**.

[0084] The normal reading posture is suitable for reading a sheet-like document **P**, that is, a document **P** which has low rigidity and is easily bent. The booklet reading posture is suitable for reading a document **P** such as a plastic card or a booklet which has high rigidity and is not easily bent.

[0085] Hereinafter, the document transport path is further described. The document **P** being fed is supported in an inclined posture by a document support unit **11** and a document support **9**. When the plurality of documents **P** are supported by the document support unit **11**, the uppermost document **P** is fed downstream by a feeding roller **14**. The document support unit **11** is formed in an upper opening and closing unit **10**. The upper opening and closing unit **10** is rotatable about a rotary shaft (not illustrated), and rotates to open and close a feeding port.

[0086] The document support **9** may be in a state of being accommodated in the upper opening and closing unit **10** and a state of being deployed from the upper opening and closing unit **10**. FIG. 1 illustrates a state of closing the upper opening and closing unit **10**, and FIG. 2 illustrates a state of opening the upper opening and closing unit **10** and deploying the document support **9**. The upper opening and closing unit **10** and the document support **9** form the first unit **3**.

[0087] Note that the scanner **1** adopts a so-called center feeding method, and center positions of the documents **P** in the X-axis direction, that is, the width direction match regardless of the sizes of the documents **P**.

[0088] The feeding roller **14** and a separation roller **15** form the first roller pair **13**.

[0089] The feeding roller **14** is provided to the second unit **4**. The feeding roller **14** gains a driving force from a transport motor **47** (see FIG. 3) to rotate. The separation roller **15** is provided at a position facing the feeding roller **14** in the first unit **3**. A rotation torque is applied to the separation roller **15** by a torque limiter **15b** (see FIG. 6). With this, double feeding of the documents **P** is prevented.

[0090] Further, in the embodiment, the feeding roller **14** is provided on the upper side with respect to the document **P** placed on the document support unit **11**, and the uppermost document **P** is first fed. Alternatively, the feeding roller **14** may be provided on the lower side with respect to the document **P** placed on the document support unit **11**, and the lowermost document **P** may first be fed.

[0091] An ultrasonic detection unit **50** is provided downstream of the first roller pair **13** in the downstream feeding path **R1**. The ultrasonic detection unit **50** is configured to include a transmission unit **50A** and a reception unit **50B** that are arranged to face each other across the downstream feeding path **R1**. A control unit **80** (see FIG. 3) can detect double feeding of the documents **P**, based on a signal transmitted from the reception unit **50B**.

[0092] Note that, in FIG. 2, positions, sizes, and shapes of the transmission unit **50A** and the reception unit **50B** are schematically illustrated, and details thereof are described later with reference to FIG. 4 and the subsequent figures.

[0093] Note that, in the embodiment, the transmission unit **50A** is arranged on the lower side of the downstream feeding path **R1**, and the reception unit **50B** is arranged on the upper side of the downstream feeding path **R1**. However, the embodiment is not limited thereto. Alternatively, the reception unit **50B** may be arranged on the lower side of the downstream feeding path **R1**, and the transmission unit **50A** may be arranged on the upper side of the downstream feeding path **R1**.

[0094] The second roller pair **16** is provided downstream of the feeding roller **14** and the separation

roller **15**. The second roller pair **16** is configured by a second lower roller **17** provided to the first unit **3** and a second upper roller **18** provided to the second unit **4**. The second upper roller **18** is provided to advance and retract with respect to the second lower roller **17**, and is pressed toward the second lower roller **17** by a pressing member (omitted in illustration), for example, a coil spring. With this, the second upper roller **18** advances and retracts with respect to the second lower roller **17** according to the thickness of the document P being transported. The second lower roller **17** and the second upper roller **18** both gain a driving force from the transport motor **47** (see FIG. **3**) to rotate.

[0095] When the second unit **4** is closed with respect to the first unit **3**, the second lower roller **17** and the second upper roller **18** contact with each other. When the second unit **4** is opened with respect to the first unit **3**, the second upper roller **18** is separated from the second lower roller **17**.

[0096] The first reading unit **32** and the second reading unit **33** are arranged to face each other downstream of the second roller pair **16**. The first reading unit **32** is provided to the first unit **3**, and the second reading unit **33** is provided to the second unit **4**.

[0097] The first reading unit **32** reads the lower surface, that is, the first surface S1 of the document P supported on the document support unit **11**, and the second reading unit **33** reads the upper surface, that is, the second surface S2 of the document P supported on the document support unit **11**. The second reading unit **33** is provided to advance and retract with respect to the first reading unit **32**, and is pressed toward the first reading unit **32** by a pressing spring **34** being an example of a pressing member. With this, the second reading unit **33** advances and retracts with respect to the first reading unit **32** according to the thickness of the document P being transported.

[0098] In the embodiment, the first reading unit **32** and the second reading unit **33** is configured by a contact image sensor module (CISM).

[0099] The third roller pair **20** is provided downstream of the first reading unit **32** and the second reading unit **33**. The third roller pair **20** is configured by a third lower roller **21** provided to the first unit **3** and a third upper roller **22** provided to the second unit **4**. The third upper roller **22** is provided to advance and retract with respect to the third lower roller **21**, and is pressed toward the third lower roller **21** by a pressing member (omitted in illustration), for example, a coil spring.

[0100] The third lower roller **21** and the third upper roller **22** both gain a driving force from the transport motor **47** (see FIG. **3**) to rotate.

[0101] When the second unit **4** is closed with respect to the first unit **3**, the third lower roller **21** and the third upper roller **22** contact with each other. When the second unit **4** is opened with respect to the first unit **3**, the third upper roller **22** is separated from the third lower roller **21**.

[0102] The flap **35** is provided downstream of the third roller pair **20**. The flap **35** rotates to switch the document transport paths described above.

[0103] Note that, in the embodiment, the flap **35** is configured to rotate in conjunction with switching of the posture of the device main body **2**. In the embodiment, as a configuration for rotating the flap **35** in conjunction with switching of the posture of the device main body **2**, mechanical rotation is performed by a conjunction mechanism (omitted in illustration), for example, a cam mechanism in conjunction with the posture of the device main body **2**. However, the flap **35** may rotate by a solenoid (omitted in illustration). In such a case, the control unit **80** (see FIG. **3**) that executes various controls drives the solenoid to rotate the flap **35**, based on detection information of a posture detection sensor (omitted in illustration).

[0104] The U-turn discharge path R3 is provided with a fourth roller pair **24** and a fifth roller pair **28**.

[0105] The fourth roller pair **24** is configured by a fourth driving roller **25** provided to the third unit **5** and a fourth driven roller **26** provided to the second unit **4**. The fourth driven roller **26** is provided to advance and retract with respect to the fourth driving roller **25**, and is pressed toward the fourth driving roller **25** by a pressing member (omitted in illustration), for example, a coil spring. With this, the fourth driven roller **26** advances and retracts with respect to the fourth driving roller **25**

according to the thickness of the document P being transported. The fourth driving roller **25** is driven by the transport motor **47** (see FIG. 3). The fourth driven roller **26** is a roller that is driven to rotate.

[0106] The fifth roller pair **28** is configured by a fifth driving roller **29** provided to the third unit **5** and a fifth driven roller **30** provided to the second unit **4**. The fifth driven roller **30** is provided to advance and retract with respect to the fifth driving roller **29**, and is pressed toward the fifth driving roller **29** by a pressing member (omitted in illustration), for example, a coil spring. With this, the fifth driven roller **30** advances and retracts with respect to the fifth driving roller **29** according to the thickness of the document P being transported. The fifth driving roller **29** is driven by the transport motor **47** (see FIG. 3). The fifth driven roller **30** is a roller that is driven to rotate.

[0107] When the third unit **5** is closed with respect to the second unit **4**, the fourth driving roller **25** and the fourth driven roller **26** contact with each other, and the fifth driving roller **29** and the fifth driven roller **30** also contact with each other. When the third unit **5** is opened with respect to the second unit **4**, the fourth driving roller **25** and the fourth driven roller **26** are separated from each other, and the fifth driving roller **29** and the fifth driven roller **30** are also separated from each other.

[0108] The document P discharged from the U-turn discharge path R3 is discharged in an obliquely upward direction containing a -Y-direction component by the fifth roller pair **28**, and is supported in an inclined posture by an upper surface **4a** of the second unit **4**.

[0109] Note that, in FIG. 2, the direction indicated by the arrow Da1 indicates a normal line direction with respect to the first surface S1 of the document P and the second surface S2 opposite thereto at a document nipping position of the first roller pair **13**. Note that the first surface S1 of the document P and the second surface S2 opposite thereto are collectively referred to as a document surface S when the surfaces are not distinguished from each other. This is because the surfaces are parallel to each other in most cases.

[0110] Further, the arrow Da2 indicates a normal direction with respect to the document surface S of the document P passing through between the transmission unit **50A** and the reception unit **50B** of the ultrasonic detection unit **50**.

[0111] Further, the arrow Da3 indicates a normal direction with respect to the document surface S at a document nipping position of the second roller pair **16**.

[0112] Further, the arrow Da4 indicates a normal direction with respect to the document surface S of the document P passing through between the first reading unit **32** and the second reading unit **33**.

[0113] Further, the arrow Da5 indicates a normal direction with respect to the document surface S at a document nipping position of the third roller pair **20**.

[0114] In the embodiment, differences between normal directions Da1, Da2, Da3, Da4, and Da5 are slight. As an example, a difference between the maximum angle and the minimum angle is less than 5 degrees. Thus, in the following description, when the respective normal line directions are not distinguished from one another, the normal line directions are referred to as a normal direction Da.

[0115] Further, in FIG. 2, the direction indicated by the arrow Fp indicates a direction that is orthogonal to the normal direction Da, and indicates a direction in which the document P is transported. Hereinafter, the direction is referred to as a transport direction Fp.

[0116] Note that, in FIG. 4 and the subsequent figures, it is assumed that an X-Fp-Da coordinate system is used in some cases.

Control System of Scanner

[0117] Next, with reference to FIG. 3, a control system of the scanner **1** is described.

[0118] The control unit **80** includes an arithmetic unit **81** configured by a central processing unit (CPU) and a storage unit **85** configured by a non-volatile memory or a volatile memory.

[0119] The control unit **80** is coupled to the first reading unit **32**, the second reading unit **33**, the transport motor **47**, and the ultrasonic detection unit **50**, and the control unit **80** controls those

components. Note that the transport motor **47** is a driving source of the feeding roller **14**, the second lower roller **17**, the second upper roller **18**, the third lower roller **21**, the third upper roller **22**, the fourth driving roller **25**, and the fifth driving roller **29**.

[0120] The control unit **80** is coupled to an interface unit **86**, receives various data or a signal that is input from an external device **87** such as a personal computer, and outputs read data that is read by the scanner **1** to the external device **87**.

[0121] Various data for controlling the scanner **1** and various programs are recorded in the storage unit **85**.

[0122] The arithmetic unit **81** reads and executes the various programs stored in the storage unit **85** to function as a transport control unit **82**, a reading control unit **83**, a double-feeding determination unit **84**, and the like.

[0123] The transport control unit **82** controls the transport motor **47** to rotate the plurality of rollers described above. With this, the document P is fed, transported, and discharged.

[0124] The reading control unit **83** controls the first reading unit **32** and the second reading unit **33** to read an image of the document P while the document P is transported.

[0125] The double-feeding determination unit **84** is a state detection unit that detects a state of the document P. In the embodiment, the double-feeding determination unit **84** determines whether double feeding of the documents P occurs, based on a reception signal that is input from a reception circuit **59** (described later).

[0126] Specifically, when a voltage value of the reception signal is less than a predetermined threshold value, it is determined that double-feeding of the documents P occurs. Note that, when the double-feeding determination unit **84** determines that double-feeding occurs, the transport control unit **82** stops the transport of the document P.

Basic Configuration of Ultrasonic Detection Unit

[0127] Next, a basic configuration of the ultrasonic detection unit **50** is described.

[0128] In FIG. 22, the ultrasonic detection unit **50** is configured to include a pair of ultrasonic elements. One of the pair of ultrasonic elements is a transmission sensor chip **53**, and the transmission sensor chip **53** transmits an ultrasonic wave. The other of the pair of ultrasonic elements is a reception sensor chip **56**, and the reception sensor chip **56** receives an ultrasonic wave.

[0129] The transmission sensor chip **53** and the reception sensor chip **56** face each other on a sensor center axis Lx being an example of a first axis, and are arranged across the document P that is transported in the downstream feeding path R1.

[0130] In the ultrasonic detection unit **50**, the transmission sensor chip **53** emits an ultrasonic wave toward the first surface S1 of the document P being fed. The ultrasonic wave emitted from the transmission sensor chip **53** passes through the document P. The ultrasonic wave transmitted through the document P is received by the reception sensor chip **56**. When the ultrasonic wave is received by the reception sensor chip **56**, a reception signal corresponding to a sound pressure of the received ultrasonic wave is output, and it is determined whether double feeding of the documents P occurs, based on the signal intensity of the reception signal.

[0131] The sensor center axis Lx is an axis passing through the center of the transmission sensor chip **53** and the center of the reception sensor chip **56**, and is a transmission/reception direction of an ultrasonic wave. The sensor center axis Lx is orthogonal to a transmission surface **53a** being an upper surface of the transmission sensor chip **53** and a reception surface **56a** being an upper surface of the reception sensor chip **56**.

[0132] Further, the sensor center axis Lx is inclined at an angle β with respect to the document surface S of the document P.

[0133] Here, when the sensor center axis Lx matches with the normal line direction of the document surface S, in other words, when the angle β is 90 degrees, an ultrasonic wave transmitted from the transmission sensor chip **53** may be reflected for a plurality of times between the

document P and the transmission sensor chip 53. Further, an ultrasonic wave passing through the document P may be reflected for a plurality of times between the reception sensor chip 56 and the document P. In such a case, in the reception sensor chip 56, in addition to the ultrasonic wave that is emitted from the transmission sensor chip 53, passes through the document P, and is received by the reception sensor chip 56, the ultrasonic wave that is reflected for a plurality of times between the document P and the transmission sensor chip 53 and the ultrasonic wave that is reflected for a plurality of times between the reception sensor chip 56 and the document P are received by the reception sensor chip 56. Thus, double-feeding cannot be detected accurately.

[0134] In contrast, the sensor center axis Lx is inclined with respect to the normal line of the document surface S of the document P. With this, reception of unnecessary ultrasonic wave components such as an ultrasonic wave that is reflected for a plurality of times can be reduced, and double feeding can be detected at high accuracy.

[0135] Note that the angle β may be set from 60 degrees to 80 degrees, and is set to 70 degrees in the embodiment.

[0136] Note that, when the area of the transmission surface 53a of the transmission sensor chip 53 is small, an ultrasonic beam diameter is reduced. In such a case, in a case in which a distance between the transmission sensor chip 53 and the reception sensor chip 56, in other words, a distance along the sensor center axis Lx is small, when the transmission sensor chip 53 is deviated from the sensor center axis Lx, a sound pressure is reduced significantly due to an attachment error. Similarly, when the area of the reception surface 56a of the reception sensor chip 56 is small, a sound pressure is reduced significantly during deviation from the sensor center axis Lx. Therefore, the distance between distance between the transmission sensor chip 53 and the reception sensor chip 56 may be secured at a certain degree. However, when the distance between the transmission sensor chip 53 and the reception sensor chip 56 is excessively increased, a sound pressure is also reduced. In the embodiment, in view of the above-mentioned point, the distance along the sensor center axis Lx between the transmission sensor chip 53 and the reception sensor chip 56, in other words, the distance between the sensors is set to a range from 15.0 mm to 30.0 mm, and is set to 24.9 mm as an example.

Configuration of Sensor Chip

[0137] A configuration of the ultrasonic detection unit 50 is further described.

[0138] As illustrated in FIG. 3, the transmission substrate 52 is provided with a transmission circuit 58 that controls the transmission sensor chip 53, and a reception substrate 55 is provided with the reception circuit 59 that controls the reception sensor chip 56.

[0139] Here, with reference to FIG. 21, a configuration of the transmission sensor chip 53 is described.

[0140] The transmission sensor chip 53 includes a base substrate 213, an element substrate 210, and a piezoelectric element 220.

[0141] The element substrate 210 includes a substrate main body portion 211 and a vibration plate 212 provided to one surface side of the substrate main body portion 211. Here, in the following description, a substrate thickness direction of the element substrate 210 is referred to as a G direction. The G direction is a direction in which an ultrasonic wave is transmitted, and is parallel to the sensor center axis Lx.

[0142] The substrate main body portion 211 is a substrate provided at the vibration plate 212, and is configured by a semiconductor substrate such as Si. An opening portion 211A that passes through the substrate main body portion 211 along the G direction is provided in the substrate main body portion 211.

[0143] The vibration plate 212 is supported by the base substrate 213. The vibration plate 212 and the base substrate 213 are bonded and fixed to each other. A space for arranging the piezoelectric element 220 is provided between the base substrate 213 and the vibration plate 212.

[0144] Note that the vibration plate 212 may be fixed while being stacked on the base substrate

213.

[0145] The vibration plate **212** is configured by SiO₂, a laminate body of SiO₂ and GrO₂, or the like, and is provided on the -G side of the substrate main body portion **211**. The vibration plate **212** closes the -G side of the opening portion **211A**. A portion of the vibration plate **212** that overlaps with the opening portion **211A** as viewed in the G direction forms a vibration portion **212A**.

[0146] The piezoelectric element **220** is provided at a position on the vibration plate **212** so as to overlap with each vibration portion **212A** as viewed in the G direction. The piezoelectric element **220** is configured by stacking a first electrode **221**, a piezoelectric film **222**, and a second electrode **223** sequentially on the vibration plate **212**.

[0147] Here, one vibration portion **212A** and the piezoelectric element **220** provided on the vibration portion **212A** form one ultrasonic transducer Tr. In the embodiment, although omitted in illustration, such ultrasonic transducers Tr are arranged in a two-dimensional array structure to form the transmission sensor chip **53**. The transmission circuit **58** (see FIG. 3) is electrically coupled to each ultrasonic transducer Tr of the transmission sensor chip **53**, and generates a driving signal for driving each ultrasonic transducer Tr.

[0148] In the transmission sensor chip **53**, a pulse wave voltage having a predetermined frequency is applied between the first electrode **221** and the second electrode **223** of each of the ultrasonic transducers Tr. With this, the piezoelectric film **222** expands and contracts. With this, the vibration portion **212A** vibrates at a frequency corresponding to an opening width of the opening portion **211A** or the like, and an ultrasonic wave is transmitted from the vibration portion **212A** toward the +G side along the sensor center axis Lx. In other words, the +G-side surface of the element substrate **210** is the ultrasonic transmission surface **53a** of the transmission sensor chip **53**.

[0149] Note that the base substrate **213** of the transmission sensor chip **53** is bonded to the transmission substrate **52** via a resist by a non-conductive adhesive. Further, at least one of a chip inclination prevention pat (omitted in illustration) for preventing inclination of the transmission sensor chip **53** with respect to the transmission substrate **52**, a positioning mark portion (omitted in illustration) for positioning the transmission sensor chip **53** with respect to the transmission substrate **52**, and an adhesive leakage prevention portion (omitted in illustration) for preventing leakage of a non-conductive adhesive is provided at the transmission substrate **52**. Further, the transmission sensor chip **53** is bonded to the transmission substrate **52** by silver paste at a position different from the chip inclination prevention pat (omitted in illustration), and the chip inclination prevention pat (omitted in illustration) is not used for conduction.

[0150] Note that, although a configuration of the reception sensor chip **56** is omitted in illustration, the reception sensor chip **56** has a configuration similar to that of the transmission sensor chip **53**. In such a case, the +G-side surface of the element substrate **210** is the ultrasonic reception surface **56a** of the reception sensor chip **56**. When an ultrasonic wave arrives at the ultrasonic transducer Tr, the vibration plate **212** vibrates according to the sound pressure of the ultrasonic wave. When the vibration plate **212** vibrates, the piezoelectric film **222** is deformed, and a potential difference is caused between the first electrode **221** and the second electrode **223**. With this, the first electrode **221** of the ultrasonic transducer Tr outputs a reception signal corresponding to the sound pressure of the received ultrasonic wave. In other words, the ultrasonic wave is detected.

[0151] As the reception circuit **59** (FIG. 3) that processes the reception signal described above, a general circuit that processes a reception signal being input through reception of an ultrasonic wave may be used. For example, the reception circuit **59** may be configured by a band-pass filter, an amplifier, a sample-and-hold circuit, a comparator, or the like (omitted in illustration). Therefore, the reception circuit **59** may also be referred to as an amplifier in view of an amplification function.

[0152] Note that the ultrasonic detection unit **50** according to the embodiment includes a voltage application portion **51** that applies a bias voltage to the reception sensor chip **56**. The voltage application portion **51** may be mounted on a dedicated substrate, may be mounted on the reception substrate **55**, or may be a part of the control unit **80**.

[0153] Here, when the ultrasonic detection unit **50** is calibrated, the reception sensor chip **56** receives an ultrasonic wave while a bias voltage is applied to the reception sensor chip **56**, and the reception circuit **59** amplifies the reception signal. Calibration is performed by emitting an ultrasonic wave from the transmission sensor chip **53** without the document P. Based on the sound pressure of the ultrasonic wave received by the reception sensor chip **56** during the calibration, an output of an ultrasonic wave to be transmitted is adjusted.

[0154] When the piezoelectric element **220** is used in the reception sensor chip **56**, a bias voltage is applied to the reception sensor chip **56**. With this, variations in the polarization direction of the piezoelectric element **220** are less likely to occur, an efficient amplitude can be obtained, and an appropriate reception signal can be output.

[0155] Similarly, when the ultrasonic detection unit **50** detects double feeding of the documents P, the reception sensor chip **56** receives an ultrasonic wave while a bias voltage is applied to the reception sensor chip **56**, and the reception circuit **59** amplifies the reception signal. With this, similarly, when double feeding of the documents P is detected, variations in the polarization direction of the piezoelectric element **220** are less likely to occur, an efficient amplitude can be obtained, and an appropriate reception signal can be output.

Configurations of Transmission Sensor Unit and Reception Sensor Unit

[0156] Next, configurations of the transmission unit **50A** and the reception unit **50B** are described.

[0157] FIG. **4** illustrates a state in which the second unit **4** is opened with respect to the first unit **3**. When the second unit **4** is opened with respect to the first unit **3**, the inside of the first unit **3** and the inside of the second unit **4** are exposed.

[0158] A transmission-side path formation member **36** is provided to the inside of the first unit **3**. When the second unit **4** is opened with respect to the first unit **3**, the transmission-side path formation member **36** is exposed.

[0159] Further, a reception-side path formation member **39** is provided to the inside of the second unit **4**. When the second unit **4** is opened with respect to the first unit **3**, the reception-side path formation member **39** is exposed.

[0160] The transmission-side path formation member **36** and the reception-side path formation member **39** form a part of the upstream feeding path **R0**, the downstream feeding path **R1**, and the reading transport path **R2**.

[0161] As illustrated in FIG. **5**, the transmission-side path formation member **36** is provided with a cover **37**. The cover **37** is provided to be attachable to and removable from the transmission-side path formation member **36** as in an attached state illustrated in FIG. **5** and FIG. **6** and a removed state illustrated in FIG. **7**. The cover **37** is provided to be attachable to the transmission-side path formation member **36** by a snap-fit structure (omitted in illustration).

[0162] An opening **37a** having a circular shape is formed in the cover **37**, and is configured so that the sensor center axis **Lx** of the ultrasonic detection unit **50** passes through the vicinity of the center of the opening **37a**. A part of a first protection member **70**, which is described later, is exposed on the inside of the opening **37a**.

[0163] The transmission unit **50A** is provided on the inside of provided the cover **37**. As illustrated in FIG. **7**, when the cover **37** is removed, the transmission unit **50A** is exposed. Note that, as illustrated in FIG. **6**, when the cover **37** is attached, a part of the transmission unit **50A** is also exposed. When the cover **37** is removed, the transmission unit **50A** is further largely exposed. When the cover **37** is attached as illustrated in FIG. **6**, a part of a first wall portion **60b** of a transmission substrate holder **60**, which is described later, is exposed.

[0164] Note that a roller holding member **38** that holds the separation roller **15** is provided to the inside of the cover **37**. When the cover **37** is removed, the roller holding member **38** can be removed, and the separation roller **15** can be replaced together with the roller holding member **38**.

[0165] Note that, in FIG. **6**, the reference symbol **15a** is a rotary shaft of the separation roller **15**, and the reference symbol **15b** is a torque limiter that applies a separation torque to the separation

roller **15**.

[0166] Note that, in FIG. 5, the reference symbol **Ba1** is an inter-axial distance between the rotary shaft center of the separation roller **15** and the rotary shaft center of the second lower roller **17**, in other words, an inter-axial distance, and the reference symbol **Ba2** is a distance between the rotary shaft center of the second lower roller **17** and the rotary shaft center of the third lower roller **21**, in other words, an inter-axial distance.

[0167] In the embodiment, the inter-axial distance **Ba1** is shorter than the inter-axial distance **Ba2**. In other words, the path length of the downstream feeding path **R1** is shorter than the path length of the reading transport path **R2**.

[0168] Note that, in the embodiment, the inter-axial distance **Ba1** is 29.1 mm, and the inter-axial distance **Ba2** is 34.0 mm.

[0169] Next, as illustrated in FIG. 8, the transport motor **47** and a main substrate **48** are provided on a side of the transmission-side path formation member **36** in the $-Da$ direction. The transport motor **47** is provided at the end of the transmission-side path formation member **36** in the $-X$ direction on the back surface side.

[0170] The main substrate **48** forms the control unit **80** described above. As illustrated in FIG. 9, the transmission unit **50A** is provided between the main substrate **48** and the transmission-side path formation member **36**.

[0171] A configuration of the transmission unit **50A** is described later in detail. The transmission unit **50A** includes the transmission substrate holder **60** as a base body, and the transmission substrate holder **60** is fixed to the transmission-side path formation member **36** by screws **61** and **61**.

[0172] Next, as illustrated in FIG. 10, an opening **39a** having a circular shape is formed in the reception-side path formation member **39**, and is configured so that the sensor center axis Lx of the ultrasonic detection unit **50** passes through the vicinity of the center of the opening **39a**. A part of a second protection member **73**, which is described later, is exposed on the inside of the opening **39a**. Note that, in FIG. 10, the reference symbol **Bb1** is a distance between the rotary shaft center of the feeding roller **14** and the rotary shaft center of the second upper roller **18**, in other words, an inter-axial distance, and the reference symbol **Bb2** is a distance between the rotary shaft center of the second upper roller **18** and the rotary shaft center of the third upper roller **22**, in other words, an inter-axial distance.

[0173] In the embodiment, the inter-axial distance **Bb1** is shorter than the inter-axial distance **Bb2**. In other words, the path length of the downstream feeding path **R1** is shorter than the path length of the reading transport path **R2**.

[0174] Note that, in the embodiment, the inter-axial distance **Bb1** is 30.0 mm, and the inter-axial distance **Bb2** is 34.0 mm.

[0175] As illustrated in FIG. 11, the reception unit **50B** is provided on a side of the reception-side path formation member **39** in the $+Da$ direction.

[0176] A configuration of the reception unit **50B** is described later in detail. The reception unit **50B** includes a reception substrate holder **62** as a base body, and a fixed portion **62j** being a portion forming the reception substrate holder **62** is fixed to the reception-side path formation member **39** by a reception-side holder fixing screw **63**. Note that the reference symbol **39b** is a positioning portion provided to the reception-side path formation member **39**. When the positioning portion **39b** is fitted into a hole portion **62e** formed in a positioned portion **62k** of the reception substrate holder **62**, the positioned portion **62k** is positioned with respect to the reception-side path formation member **39**.

[0177] In this manner, the reception substrate holder **62** is positioned and fixed with respect to the reception-side path formation member **39**.

[0178] Next, configurations of the transmission unit **50A** and the reception unit **50B** are further described with reference to FIG. 12 and the subsequent figures.

[0179] As illustrated in FIG. 12, the ultrasonic detection unit **50** is arranged in a space between the first roller pair **13** and the second roller pair **16**. Along the transport direction F_p , the separation roller **15**, the transmission unit **50A** forming the first roller pair **13**, the second lower roller **17** forming the second roller pair **16**, and the third lower roller **21** forming the third roller pair **20** are arranged in the stated order. Further, along the transport direction F_p , the feeding roller **14** forming the first roller pair **13**, the reception unit **50B**, the second upper roller **18** forming the second roller pair **16**, and the third upper roller **22** forming the third roller pair **20** are arranged in the stated order.

[0180] As illustrated in FIG. 13 and FIG. 14, the transmission unit **50A** is configured by providing the transmission substrate **52** and the first protection member **70** to the transmission substrate holder **60**.

[0181] Similarly, the reception unit **50B** is configured by providing the reception substrate **55** and the second protection member **73** to the reception substrate holder **62**.

[0182] As illustrated in FIG. 17 and FIG. 18, the transmission substrate holder **60** includes screw holes **60e** respectively for screw fixing portions **60j** and **60k**, and the screws **61**, which are described with reference to FIG. 9, pass through the screw holes **60e**. A fixing portion **60a** is formed to be positioned in the +Da direction with respect to the screw fixing portions **60j** and **60j** between the screw fixing portion **60j** and the screw fixing portion **60k**. The fixing portion **60a** is a portion for fixing the first protection member **70**, and a first opening **60f** is formed at the center portion (see FIG. 17 and FIG. 18 and FIG. 22). The first opening **60f** is formed so that the sensor center axis L_x of the ultrasonic detection unit **50** passes through the vicinity of the center of the first opening **60f**.

[0183] Protection member positioning portions **60c** and **60c** are formed on the surface of the fixing portion **60a**. As illustrated in FIG. 19, the first protection member **70** is held by a first holding member **71**, and recessed portions **71a** and **71a** are formed in the first holding member **71**. When the protection member positioning portions **60c** and **60c** are fitted into the recessed portions **71a** and **71a**, the first holding member **71**, in other words, the first protection member **70** is positioned with respect to the fixing portion **60a**.

[0184] As illustrated in FIG. 18, a protrusion portion **60m** having a cylindrical shape is formed on the back side of the fixing portion **60a**, and a screw hole **60g** is formed in the protrusion portion **60m**. Further, as illustrated in FIG. 14, a substrate positioning portion **60d** is formed on the back side of the fixing portion **60a**.

[0185] As illustrated in FIG. 15, an opening **52a** and a recessed portion **52b** are formed in the transmission substrate **52**. The protrusion portion **60m** of the transmission substrate holder **60** is fitted into the opening **52a**, and the substrate positioning portion **60d** of the transmission substrate holder **60** is fitted into the recessed portion **52b**. With this, the transmission substrate **52** is positioned with respect to the transmission substrate holder **60**. Further, as illustrated in FIG. 14, a transmission substrate fixing screw **54** is fitted into the screw hole **60g** (FIG. 18). In this manner, the transmission substrate **52** is fixed with respect to the transmission substrate holder **60**.

[0186] The transmission substrate holder **60** holds the transmission substrate **52** so that the transmission surface **53a** of the transmission sensor chip **53** is inclined at an angle α_1 with respect to the document surface S as illustrated in FIG. 22. With this, the transmission surface **53a** of the transmission sensor chip **53** and the reception surface **56a** of the reception sensor chip **56** are parallel to each other, and the sensor center axis L_x is inclined at the angle β with respect to the document surface S .

[0187] Note that, as illustrated in FIG. 13, FIG. 14, FIG. 17, and FIG. 18, the first wall portion **60b** is formed upstream of the fixing portion **60a** in the transport direction F_p , and a second wall portion **60h** is formed downstream of the fixing portion **60a** in the transport direction F_p .

[0188] As illustrated in FIG. 6, the first wall portion **60b** is at a position so that the first wall portion **60b** can be seen from the outside while the cover **37** is closed, and includes a function of

protecting the transmission substrate **52** from a foreign matter, a liquid, or the like.

[0189] Although omitted in illustration, the first wall portion **60b** is at a position overlapping with a part of the transmission substrate **52** and the entire transmission sensor chip **53** as viewed from upstream in the transport direction F_p . In other words, the first wall portion **60b** is at such a position that a part of the transmission substrate **52** and the entire transmission sensor chip **53** are hidden by the first wall portion **60b** as viewed from upstream in the transport direction F_p .

[0190] Next, the reception substrate holder **62** includes the fixed portion **62j** and the positioned portion **62k** which are described with reference to FIG. **11**, and a fixing portion **62a** is formed between the fixed portion **62j** and the positioned portion **62k** so as to be positioned in the $-D_a$ direction with respect to the fixed portion **62j** and the positioned portion **62k**. The fixing portion **62a** is a portion for fixing the second protection member **73**, and a second opening **62f** is formed at the center portion (see FIG. **22**). The second opening **62f** is formed so that the sensor center axis L_x of the ultrasonic detection unit **50** passes through the vicinity of the center of the second opening **62f**.

[0191] As illustrated in FIG. **14**, protection member positioning portions **62c** and **62c** are formed on the surface of the fixing portion **62a**. The second protection member **73** is held by a second holding member **74**, and recessed portions **74a** and **74a** are formed in the second holding member **74**. When the protection member positioning portions **62c** and **62c** are fitted into the recessed portions **74a** and **74a**, the second holding member **74**, in other words, the second protection member **73** is positioned with respect to the fixing portion **62a**.

[0192] As illustrated in FIG. **13**, a substrate positioning portion **62d** having a protrusion shape is formed on the back side of the fixing portion **62a**. A recessed portion **55b** is formed in the reception substrate **55**. The substrate positioning portion **62d** is fitted into the recessed portion **55b**. Further, although omitted in illustration, a protrusion portion and a screw hole which are similar to the protrusion portion **60m** and the screw hole **60g** of the transmission substrate holder **60**, respectively, are formed at the reception substrate holder **62**. Further, an opening similar to the opening **52a** of the transmission substrate **52** (see FIG. **15**) is formed in the reception substrate **55**. Further, the protrusion portion of the reception substrate holder **62** is fitted into the opening, and the substrate positioning portion **62d** of the reception substrate holder **62** is fitted into the recessed portion **55b** of the reception substrate **55**. With this, the reception substrate **55** is positioned with respect to the reception substrate holder **62**. Further, as illustrated in FIG. **13**, a reception substrate fixing screw **57** is fitted into the screw hole described above. With this, the reception substrate **55** is fixed with respect to the reception substrate holder **62**.

[0193] The reception substrate holder **62** holds the reception substrate **55** so that the reception surface **56a** of the reception sensor chip **56** is inclined at the angle α_1 with respect to the document surface S as illustrated in FIG. **22**. With this, the reception surface **56a** of the reception sensor chip **56** and the transmission surface **53a** of the transmission sensor chip **53** are parallel to each other, and the sensor center axis L_x is inclined at the angle β with respect to the document surface S .

[0194] As described above, in the embodiment, the transmission substrate **52** is fixed to the transmission-side path formation member **36**, which is a member forming the document transport path, via the transmission substrate holder **60**, and the reception substrate **55** is fixed to the reception-side path formation member **39**, which is a member forming the document transport path, via the reception substrate holder **62**. With this, a distance between the document P and the transmission sensor chip **53** and a distance between the document P and the reception sensor chip **56** are stabilized, and a detection value of an ultrasonic wave is stabilized.

[0195] Next, a configuration of the transmission substrate **52**

[0196] is described.

[0197] As illustrated in FIG. **13** to FIG. **15**, the transmission substrate **52** is a rectangular substrate having short sides E_1 and long sides E_2 . As described above, the recessed portion **52b** is formed at one of the short sides E_1 . The transmission substrate **52** is provided so that the short side E_1

extends along the transport direction Fp.

[0198] The transmission sensor chip **53** is fixed while being placed on a substrate surface **52d** of the transmission substrate **52**. No other member, such as a pedestal, is interposed between the transmission sensor chip **53** and the substrate surface **52d**, except for an adhesive.

[0199] A connector **52c** is provided on a side opposite to the substrate surface **52d** of the transmission substrate **52**, and a transmission-side cable **76** (see FIG. **13** and FIG. **14**) is connected to the connector **52c**.

[0200] Further, the substrate surface **52d** of the transmission substrate **52** is provided with a first surrounding member **65**. In the embodiment, the first surrounding member **65** is formed of a metal plate material, and is fixed to the transmission substrate **52** by an adhesive.

[0201] An opening **65a** is formed in the first surrounding member **65**, and the opening **65a** is formed so that the sensor center axis Lx of the ultrasonic detection unit **50** passes through the vicinity of the center of the opening **65a**. The first surrounding member **65** is provided to extend along the substrate surface **52d** as illustrated in FIG. **16**.

[0202] The transmission substrate fixing screw **54** includes a head portion including a driver fitting hole, a screw portion including a male screw, and a distal end portion of a distal end of the male screw. The screw portion is attached to an attached portion on the same side of the transmission sensor chip **53**, specifically, the screw hole **60g** (see FIG. **18** and FIG. **22**) in the substrate surface **52d** of the transmission substrate **52**. Note that the screw portion may be attached to an attached portion on a side opposite to a transmission sensor chip **53**, in other words, a screw hole in the substrate surface **52d** of the transmission substrate **52**.

[0203] While the transmission substrate **52** is fixed to the transmission substrate holder **60**, a position of a distal end portion **54a** of the transmission substrate fixing screw **54** is located in the -Da direction with respect to a most +Da-direction position **65b** of the first surrounding member **65** in the normal direction Da.

[0204] Further, a direction extending along the sensor center axis Lx from the transmission unit **50A** to the reception unit **50B** is a +Lx direction, and a direction opposite thereto is a -Lx direction. While the transmission substrate **52** is fixed to the transmission substrate holder **60**, the position of the distal end portion **54a** of the transmission substrate fixing screw **54** is positioned in the +Lx direction with respect to the first surrounding member **65**.

[0205] Further, a distance from the substrate surface **52d** of the transmission substrate **52** to the first surrounding member **65** is less than a length of the screw portion of the transmission substrate fixing screw **54**.

[0206] In FIG. **16**, the reference symbol **t0** is a thickness including the transmission substrate **52** and the transmission sensor chip **53**, the reference symbol **t1** is a thickness of the transmission substrate **52**, and the reference symbol **t2** is a thickness of the transmission sensor chip **53**. The thickness **t2** of the transmission sensor chip **53** is less than the thickness of the transmission substrate **52**.

[0207] As an example, the thickness of the transmission substrate **52** may be 0.8 to 1.2 mm, in particular, 1.0 mm. Further, as an example, the thickness of the transmission sensor chip **53** may be 0.5 to 0.7 mm, in particular, 0.578 mm.

[0208] As an example, the distance from the substrate surface **52d** of the transmission substrate **52** to the first surrounding member **65** may be 1.2 to 1.6 mm, in particular, 1.4 mm.

[0209] Next, a configuration of the reception substrate **55** is described.

[0210] In the embodiment, as illustrated in FIG. **13** and FIG. **14**, the shape and the dimension of the reception substrate **55** are similar to those of the transmission substrate **52**.

[0211] In other words, the reception substrate **55** is a rectangular substrate including short sides E1 and long sides E2, and is provided so that the short side E1 extends along the transport direction Fp.

[0212] The reception substrate **55** is provided with a connector **55c**, and a reception-side cable **77** is

connected to the connector **55c**.

[0213] With reference to FIG. **30**, the configuration of the reception substrate **55** is further described below. Note that, in FIG. **30** to FIG. **32**, for the sake of illustration, the reception substrate **55** is illustrated horizontally.

[0214] The reception sensor chip **56** is fixed while being placed on a first substrate surface **55d** of the reception substrate **55d**. No other member, such as a pedestal, is interposed between the reception sensor chip **56** and the first substrate surface **55d**, except for an adhesive.

[0215] Note that the first substrate surface **55d** is a surface facing the transmission sensor chip **53**, and the reference symbol **55e** is a surface opposite to the first substrate surface **55d**. The connector **55c** is provided to the second substrate surface **55e**.

[0216] The first substrate surface **55d** is provided with a second surrounding member **67**. In the embodiment, the second surrounding member **67** is formed of a metal plate material, and is fixed to the reception substrate **55** by an adhesive.

[0217] In the second surrounding member **67**, an opening (omitted in illustration) similar to the opening **65a** on the transmission side, which is illustrated in FIG. **15**, is formed. The opening is formed so that the sensor center axis **Lx** of the ultrasonic detection unit **50** passes through the vicinity of the center of the opening. The second surrounding member **67** is provided to extend along the first substrate surface **55d**.

[0218] Next, the second substrate surface **55e** is provided with the reception circuit **59**. The reception circuit **59** is in a form of a semiconductor chip, and an application specific integrated circuit (ASIC) is used, for example. The reception circuit **59** is fixed while being placed on the second substrate surface **55e**. No other member, such as a pedestal, is interposed between the reception circuit **59** and the second substrate surface **55e**, except for an adhesive.

[0219] The reception circuit **59** is covered with a shielding member **91**. In the embodiment, similarly to the second surrounding member **67**, the shielding member **91** is formed of a metal plate material so as to obtain an electromagnetic shield effect. The shielding member **91** is fixed to the second substrate surface **55e** by an adhesive.

[0220] In FIG. **30**, the reference symbol **t10** is a thickness of the reception substrate **55**, and is similar to the thickness **t1** of the transmission substrate **52** (see FIG. **16**). The reference symbol **t11** is a height of the reception circuit **59** with respect to the second substrate surface **55e**. The reference symbol **t12** is a height of the shielding member **91** with respect to the second substrate surface **55e**.

[0221] Further, the reference symbol **t13** is a height of the reception sensor chip **56** with respect to the first substrate surface **55d**. The reference symbol **t14** is a height of the second surrounding member **67** with respect to the first substrate surface **55d**.

[0222] As an example, the thickness **t10** may be 0.8 to 1.2 mm, in particular, 1.0 mm.

[0223] Further, as an example, the height **t11** may be 0.9 to 1.3 mm, in particular, 1.1 mm.

[0224] Further, as an example, the height **t12** may be 1.8 to 2.2 mm, in particular, 2.0 mm.

[0225] Further, as an example, the height **t13** may be 0.5 to 0.7 mm, in particular, 0.578 mm.

[0226] Further, as an example, the height **t14** may be 1.2 to 1.6 mm, in particular, 1.4 mm.

[0227] In FIG. **22**, the reception substrate fixing screw **57** includes a head portion including a driver fitting hole, a screw portion including a male screw, and a distal end portion of a distal end of the male screw. The screw portion is attached to an attached portion on the same side as the reception sensor chip **56**, specifically, a screw hole **62g** in the first substrate surface **55** of the reception substrate **55d**. Note that the screw portion may be attached to an attached portion on a side opposite to the reception sensor chip **56**, in other words, a screw hole in the first substrate surface **55** of the reception substrate **55d**.

[0228] While the reception substrate **55** is fixed to the reception substrate holder **62**, a position of a distal end portion **57a** of the reception substrate fixing screw **57** is located in the +Da direction with respect to a most -Da-direction position **67b** of the second surrounding member **67** in the normal

direction Da.

[0229] Further, while the reception substrate **55** is fixed to the reception substrate holder **62**, the position of the distal end portion **57a** of the reception substrate fixing screw **57** is located in the -Lx direction with respect to the second surrounding member **67** in the axial direction of the sensor center axis Lx.

[0230] Further, a distance from the first substrate surface **55** of the reception substrate **55d** to the second surrounding member **67** is less than a length of the screw portion of the reception substrate fixing screw **57**.

[0231] As described above, the transmission substrate **52** is provided with the first surrounding member **65** that is a member surrounding the transmission sensor chip **53** and is a plate-like member provided to extend along the transmission substrate **52**. Further, the reception substrate **55** is provided with the second surrounding member **67** that is a member surrounding the reception sensor chip **56** and is a plate-like member provided to extend along the reception substrate **55**. With this, the transmission sensor chip **53** or the reception sensor chip **56** can be protected from an external force. Further, in the embodiment, the first surrounding member **65** and the second surrounding member **67** are formed of a metal material. Thus, the transmission sensor chip **53** or the reception sensor chip **56** can be protected from an electromagnetic wave.

[0232] As described above, the reception substrate **55** includes the reception circuit **59** that amplifies a reception signal of an ultrasonic wave received by the reception sensor chip **56** and the shielding member **91** that is provided at the reception substrate **55** and covers at least a part of the reception circuit **59**.

[0233] Further, the height **t13** of the reception sensor chip **56** with respect to the reception substrate **55** is lower than the height **t12** of the shielding member **91** with respect to the reception substrate **55**. With this, in a configuration in which the reception sensor chip **56** and the shielding member **91** are provided to the reception substrate **55**, a device dimension can be reduced.

[0234] Note that, in the embodiment, the shielding member **91** covers the entire reception circuit **59**, but may cover a part of the reception circuit **59**.

[0235] Further, in the embodiment, the height **t13** of the reception sensor chip **56** with respect to the reception substrate **55** is lower than the height **t11** of the reception circuit **59** with respect to the reception substrate **55**.

[0236] Further, in the embodiment, the height **t14** of the second surrounding member **67** with respect to the reception substrate **55** is lower than the height **t12** of the shielding member **91** with respect to the reception substrate **55**.

[0237] Further, in the embodiment, the height **t12** of the embodiment of the shielding member **91** with respect to the reception substrate **55** is lower than the height **t15** of the connector **55c** with respect to the reception substrate **55**.

[0238] Further, in the embodiment, in the reception substrate **55**, the reception circuit **59** and the shielding member **91** are provided to the second substrate surface **55e** opposite to the first substrate surface **55d** provided with the reception sensor chip **56**. With this, the one surface and the other surface of the reception substrate **55** can be utilized effectively, and the reception substrate **55** can easily be reduced in size. With this, a device dimension can be reduced.

[0239] Further, in the embodiment, the reception substrate **55** is provided with the connector **55c** on the second substrate surface **55e**. Further, the reception circuit **59** and the shielding member **91** are arranged on the second substrate surface **55e**, which is provided with the connector **55c**, of the reception substrate **55**.

[0240] With this, the reception circuit **59** and the shielding member **91** are arranged in a space generated while arranging the connector **55c**. Thus, a device dimension can be reduced.

[0241] Note that, in another embodiment, as illustrated in FIG. **31**, in the reception substrate **55**, the reception circuit **59** may be arranged on the first substrate surface **55d** provided with the reception sensor chip **56**. In such a case, the shielding member **91** may cover the reception sensor chip **56** in

addition to the reception circuit **59**. In this manner, the same shielding member **91** covers the reception circuit **59** and the reception sensor chip **56**. With this, a cost increase can be suppressed while reducing a device dimension.

[0242] Note that, when the reception circuit **59** and the reception sensor chip **56** are provided to the first substrate surface **55d**, the reception circuit **59** and the reception sensor chip **56** may be covered with different shielding members as illustrated in FIG. **32**.

[0243] Further, in any one of the embodiments in FIG. **30**, FIG. **31**, and FIG. **32**, the reception sensor chip **56** and the reception circuit **59** are provided to one reception substrate **55**. Alternatively, an arrangement configuration equivalent to those in FIG. **30**, FIG. **31**, and FIG. **32** may be achieved by providing the reception sensor chip **56** and the reception circuit **59** to different substrates.

Configurations of First Protection Member and Second Protection Member

[0244] Next, the first protection member **70** and the second protection member **73** are described in detail.

[0245] Note that, in the embodiment, the first protection member **70** is formed of the same material as the second protection member **73**, and the second holding member **74** that holds the second protection member **73** is formed of the same material as the first holding member **71** that holds the first protection member **70**. Thus, the first protection member **70** and the first holding member **71** are described below, and the second protection member **73** and the second holding member **74** are omitted in description.

[0246] In the embodiment, the first protection member **70** is a filter formed into a mesh-like shape. FIG. **20** illustrates a part of the first protection member **70** in an enlarged manner.

[0247] In FIG. **20**, the first protection member **70** is a filter formed into a mesh-like shape by arranging wire rods **70a** so as to cross each other. Note that FIG. **20** illustrates an example in which the wire rods **70a** intersect at 90 degrees, which is not limited thereto. The wire rods **70a** may intersect at an angle other than 90 degrees.

[0248] As the material of the wire rod **70a**, a metal material such as copper, iron, brass, and SUS, an alloy material, a synthetic resin such as nylon and polyester, or the like may be used. As an example, in the embodiment, polyester is used.

[0249] Note that the rod diameter of the wire rod **70a** may be less than a wavelength of an ultrasonic wave. With this, the disadvantage of an ultrasonic wave being diffusely reflected by the wire rod **70a** of the first protection member **70** is suppressed.

[0250] In such a first protection member **70**, an opening **70b** is formed between a pair of adjacent wire rods **70a**. The opening **70b** corresponds to a hole portion through which an ultrasonic wave passes. In order to suppress adhesion or accumulation of a foreign matter, such as paper dust, onto the transmission surface **53a** of the transmission sensor chip **53** and the reception surface **56a** of the reception sensor chip **56**, a width of the opening **70b**, in other words, an opening width W_a may be 1 mm or less.

[0251] However, in the embodiment, when the device is cleaned by using a cleaning agent, the opening width W_a is set to 30 μm or less, and is set to 22 μm , for example so as to prevent the cleaning agent from passing through the opening **70b**.

[0252] Note that, in FIGS. **12** to **14** and FIG. **19**, the first protection member **70** or the second protection member **73** is illustrated in a mesh pattern so as to clearly indicate that the first protection member **70** and the second protection member **73** are mesh-like members. However, the intersection angle and the opening width W_a of the wire rods **70a** are merely illustrative.

[0253] As described above, the first protection member **70** and the second protection member **73** are mesh-like members, and hence an ultrasonic wave can satisfactorily pass through the first protection member **70** and the second protection member **73**.

[0254] The opening width W_a in the mesh-like member is 30 μm or less. Thus, when the device is cleaned by using a cleaning agent, the cleaning agent is prevented from passing through the first protection member **70** and adhering to the transmission sensor chip **53**, or the cleaning agent is

prevented from passing through the second protection member **73** and adhering to the reception sensor chip **56**. As a result, reduction of intensity of an ultrasonic wave received by the reception sensor chip **56** and a malfunction causing erroneous detection can be suppressed.

[0255] The first protection member **70** described above is held by the first holding member **71** including an opening **71b** for exposing the first protection member **70** as illustrated in FIG. **19**. The first protection member **70** is sandwiched between the two first holding members **71**, and thus is held by the first holding members **71**. A double sided tape (omitted in illustration) is used between the first protection member **70** and the first holding member **71**. With this, the first protection member **70** and the first holding member **71** are integrated.

[0256] Note that the relationship between the second protection member **73** and the second holding member **74** is similar to the relationship between the first protection member **70** and the first holding member **71** described above.

[0257] Next, the first protection member **70** is held by the transmission substrate holder **60** so as to be inclined at an angle $\alpha 2$ with respect to the document surface S as illustrated in FIG. **22**.

Similarly, the second protection member **73** is held by the reception substrate holder **62** so as to be inclined at the angle $\alpha 2$ with respect to the document surface S. With this, an angle formed between the first protection member **70** and the second protection member **73** is 0 degrees, in other words, the first protection member **70** and the second protection member **73** are parallel to each other.

[0258] In the embodiment, the angle $\alpha 2$ is set to an angle different from the angle $\alpha 1$, specifically, the angle $\alpha 2$ is set to be smaller than the angle $\alpha 1$.

[0259] In other words, when the angle $\alpha 2$ and the angle $\alpha 1$ are set to the same angle, in other words, the transmission surface **53a** and the first protection member **70** are parallel to each other, multiple reflections between the transmission surface **53a** and the first protection member **70** may be caused, and accurate double feeding detection may not be executed. The same applies when the reception surface **56a** and the second protection member **73** are parallel to each other.

[0260] However, as described above, the angle $\alpha 2$ is set to an angle different from the angle $\alpha 1$. Thus, multiple reflections described above can be suppressed, and double feeding can be executed appropriately.

[0261] Note that, in the embodiment, as an example, the angle $\alpha 1$ is set to 20 degrees, and the angle $\alpha 2$ is set to 15 degrees.

[0262] Note that the distance between the transmission surface **53a** and the first protection member **70** is also a factor causing multiple reflections of an ultrasonic wave. In other words, when the distance between the transmission surface **53a** and the first protection member **70** is short, multiple reflections of an ultrasonic wave are caused. In the embodiment, in view of this, the distance between the transmission surface **53a** and the first protection member **70**, in other words, the distance along the sensor center axis Lx is set to 4.0 mm to 6.0 mm, and is set to 4.89 mm, for example.

[0263] The first protection member **70** described above is held by the transmission substrate holder **60** together with the transmission substrate **52**. Thus, fluctuations in the distance between the transmission sensor chip **53** and the first protection member **70** can be suppressed. As a result, multiple reflections of an ultrasonic wave between the transmission sensor chip **53** and the first protection member **70** can be suppressed appropriately.

[0264] Similarly, the second protection member **73** is held by the reception substrate holder **62** together with the reception substrate **55**. Thus, fluctuations in the distance between the reception sensor chip **56** and the second protection member **73** can be suppressed. As a result, multiple reflections of an ultrasonic wave between the reception sensor chip **56** and the second protection member **73** can be suppressed appropriately.

[0265] Further, the transmission substrate holder **60** includes the first opening **60f** through which an ultrasonic wave passes, on the sensor center axis Lx, and the first protection member **70** is provided to cover the first opening **60f** in the surface of the transmission substrate holder **60**.

[0266] Similarly, the reception substrate holder **62** includes the second opening **62f** through which an ultrasonic wave passes, on the sensor center axis Lx , and the second protection member **73** is provided to cover the second opening **62f** in the surface of the reception substrate holder **62**.

[0267] Accordingly, the following operational effects are obtained.

[0268] When a foreign matter remains adhering to the first protection member **70** or the second protection member **73**, intensity of an ultrasonic wave received by the reception sensor chip **56** may be reduced, and erroneous detection may be caused. Therefore, in such a case, replacement of the first protection member **70** or the second protection member **73** and maintenance such as cleaning are required.

[0269] In the embodiment, the first protection member **70** is provided to cover the first opening **60f** in the surface of the transmission substrate holder **60**, and hence replacement and maintenance of the first protection member **70** are facilitated.

[0270] Further, similarly, the second protection member **73** is provided to cover the second opening **62f** in the surface of the reception substrate holder **62**, and hence replacement and maintenance of the second protection member **73** are facilitated.

[0271] Further, as described with reference to FIG. **22**, the transmission sensor chip **53** and the reception sensor chip **56** are inclined at the angle $\alpha 1$ being a first angle with respect to the surface of the document P passing through between the transmission sensor chip **53** and the reception sensor chip **56**.

[0272] Further, the first protection member **70** and the second protection member **73** are inclined at the angle $\alpha 2$ being a second angle with respect to the surface of the document P passing through between the transmission sensor chip **53** and the reception sensor chip **56**.

[0273] Further, the angle $\alpha 1$ and the angle $\alpha 2$ are different from each other, and the angle $\alpha 2$ is less acute than the angle $\alpha 1$.

[0274] With this, the following actions and effects are obtained.

[0275] When the first protection member **70** and the second protection member **73** are parallel to the surface of the document P passing through between the transmission sensor chip **53** and the reception sensor chip **56**, paper dust or the like easily adheres to the first protection member **70** and the second protection member **73**. However, the first protection member **70** and the second protection member **73** are inclined at the angle $\alpha 2$ with respect to the surface of the document P passing through between the transmission sensor chip **53** and the reception sensor chip **56**. Thus, adhesion of a foreign matter such as paper dust to the first protection member **70** and the second protection member **73** can be suppressed.

[0276] Note that the transmission sensor chip **53** and the reception sensor chip **56** are inclined at the angle $\alpha 1$ with respect to the surface of the document P passing through between the transmission sensor chip **53** and the reception sensor chip **56**. Thus, multiple reflections of an ultrasonic wave between the transmission sensor chip **53** and the document P or between the reception sensor chip **56** and the document P can be suppressed.

[0277] Further, the angle $\alpha 1$ and the angle $\alpha 2$ are different from each other, and hence multiple reflections of an ultrasonic wave between the transmission sensor chip **53** and the first protection member **70** can be suppressed. Further, multiple reflections of an ultrasonic wave between the reception sensor chip **56** and the second protection member **73** can be suppressed.

[0278] Further, as described with reference to FIG. **5** to FIG. **7**, the scanner **1** includes the transmission-side path formation member **36** being a member that forms the reading transport path **R2** being a document transport path between the first roller pair **13** and the second roller pair **16** and fixes the transmission substrate holder **60**. The transmission-side path formation member **36** is provided with the cover **37** that is removable and attachable. When the cover **37** is attached, the cover **37** covers the transmission substrate holder **60**. When the cover **37** is removed, the separation roller **15** can be replaced, and the transmission substrate holder **60** is exposed. In particular, the first protection member **70** is exposed.

[0279] With this, the transmission substrate holder **60** can easily be accessed. As a result, for example, replacement and maintenance of the first protection member **70** are facilitated.

[0280] Further, the transmission substrate holder **60** includes the fixing portion **60a** that fixes the first protection member **70**. The transmission substrate **52** is fixed to the back side of the fixing portion **60a**. The fixing portion **60a** includes the first wall portion **60b** that is a wall portion facing the side of the transmission substrate **52** being fixed and covers at least a part of the transmission substrate **52**. With this, a foreign matter moves around from the side of the fixing portion **60a** toward the transmission substrate **52**, the first wall portion **60b** can prevent such movement of the foreign matter.

Relationship Between Configuration of Ultrasonic Detection Unit and Other Configurations of Scanner

[0281] Subsequently, relative positional relationships between the configuration of the ultrasonic detection unit **50** and other configurations of the scanner **1** are described.

[0282] First, with reference to FIG. **23**, the positional relationship between the ultrasonic detection unit **50** and the first roller pair **13** is described. FIG. **23** illustrates the positional relationship between the ultrasonic detection unit **50** and the first roller pair **13** as viewed from upstream in the transport direction F_p . In FIG. **23**, the feeding roller **14** and the separation roller **15** that form the first roller pair **13** are indicated by the two-dot chain line.

[0283] As illustrated, at least a part of the transmission sensor chip **53** and at least a part of the reception sensor chip **56** are on the inner side of a range $Ua1$ of the first roller pair **13** in the normal direction Da for the document surface S at the nipping position of the first roller pair **13**. With this, the device dimension in the normal direction Da can be reduced.

[0284] Note that, in the embodiment, the entire transmission sensor chip **53** and the entire reception sensor chip **56** are on the inner side of the range $Ua1$ in the normal direction Da . However, a part of the transmission sensor chip **53** and a part of the reception sensor chip **56** may be on the inner side of the range $Ua1$ in the normal direction Da . Alternatively, a part of the transmission sensor chip **53** and the entire reception sensor chip **56** may be on the inner side of the range $Ua1$ in the normal direction Da . Alternatively, the entire transmission sensor chip **53** and a part of the reception sensor chip **56** may be on the inner side of the range $Ua1$ in the normal direction Da .

[0285] Further, in the embodiment, at least a part of the reception substrate **55** and at least a part of the shielding member **91** are on the inner side of the range $Ua1$ of the first roller pair **13** in the normal direction Da for the document surface S at the nipping position of the first roller pair **13**. With this, the device dimension in the normal direction Da can be reduced.

[0286] Note that, in the embodiment, the entire reception substrate **55** and the entire shielding member **91** are on the inner side of the range $Ua1$ in the normal direction Da . However, a part of the reception substrate **55** or a part of the shielding member **91** may be on the inner side of the range $Ua1$ in the normal direction Da .

[0287] Note that the reference symbol **14a** is a rotary shaft of the feeding roller **14**, and the reference symbol **15a** is a rotary shaft of the separation roller **15**. In the embodiment, a distance between the shaft center of the rotary shaft **14a** and the shaft center of the rotary shaft **15a**, in other words, an inter-axial distance $dk1$ is 16.6 mm, which is shorter than the distance between the transmission sensor chip **53** and the reception sensor chip **56**, in other words, the inter-sensor distance of 24.9 mm. With this, an occupied region of the first roller pair **13** in a direction along the inter-axial distance $dk1$ can be reduced, and a device dimension can be reduced.

[0288] Note that, in the embodiment, the shaft diameter of the rotary shaft **14a** is 6.0 mm, and the shaft diameter of the rotary shaft **15a** is 3.0 to 6.0 mm, for example, 3.9 mm or 5.18 mm. All of the shaft diameters thereof are greater than the thickness of the transmission sensor chip **53** or the reception sensor chip **56**, which is 0.578 mm.

[0289] Next, with reference to FIG. **24**, the positional relationship between the ultrasonic detection unit **50**, and the second roller pair **16** and the third roller pair **20** is described. FIG. **24** illustrates the

positional relationship between the ultrasonic detection unit **50**, and the second roller pair **16** and the third roller pair **20** as viewed from upstream in the transport direction F_p .

[0290] In FIG. **24**, the second lower roller **17** and the second upper roller **18** that form the second roller pair **16** are indicated by the two-dot chain line. Note that the outline of the third lower roller **21** forming the third roller pair **20** overlaps with the second lower roller **17**. Further, the outline of the third upper roller **22** forming the third roller pair **20** overlaps with the second upper roller **18**.

[0291] Note that, while details thereof are described later, both the second upper roller **18** and the third upper roller **22** are provided to advance and retract with respect to the facing rollers. FIG. **24** illustrates a state in which both the second upper roller **18** and the third upper roller **22** contact with the facing rollers.

[0292] As illustrated, at least a part of the transmission sensor chip **53** and at least a part of the reception sensor chip **56** are on the inner side of a range $Ua2$ of the second roller pair **16** or the third roller pair **20** in the normal direction Da for the document surface S at the nipping position of the second roller pair **16** or the third roller pair **20**. With this, a device dimension can be reduced in the normal direction Da for the document surface S at the nipping position of the second roller pair **16** or the third roller pair **20**.

[0293] Note that, in the embodiment, the entire transmission sensor chip **53** and the entire reception sensor chip **56** are on the inner side of the range $Ua2$ in the normal direction Da . However, a part of the transmission sensor chip **53** and a part of the reception sensor chip **56** may be on the inner side of the range $Ua2$ in the normal direction Da . Alternatively, a part of the transmission sensor chip **53** and the entire reception sensor chip **56** may be on the inner side of the range $Ua2$ in the normal direction Da . Alternatively, the entire transmission sensor chip **53** and a part of the reception sensor chip **56** may be on the inner side of the range $Ua2$ in the normal direction Da .

[0294] Note that the relationships between the dimensions of the rotary shafts of the second lower roller **17**, the second upper roller **18**, the third lower roller **21**, and the third upper roller **22** and the dimensions of the transmission sensor chip **53** and the reception sensor chip **56** are further described later.

[0295] Note that in the embodiment, a part of the transmission substrate **52** and the entire reception substrate **55** are on the inner side of the range $Ua1$ in the normal direction Da . However, the entire transmission substrate **52** and the entire reception substrate **55** may be on the inner side of the range $Ua1$ in the normal direction Da . Alternatively, a part of the transmission substrate **52** and a part of the reception substrate **55** may be on the inner side of the range $Ua1$ in the normal direction Da . Alternatively, the entire transmission substrate **52** and a part of the reception substrate **55** may be on the inner side of the range $Ua1$ in the normal direction Da .

[0296] Note that a distance between the rotary shaft center of the second lower roller **17** and the rotary shaft center of the second upper roller **18**, or a distance between the rotary shaft center of the third lower roller **21** and the rotary shaft center of the third upper roller **22**, in other words, an inter-axial distance $dk2$ is 15.4 mm, which is shorter than the distance between the transmission sensor chip **53** and the reception sensor chip **56**, in other words, the inter-center distance of 24.9 mm.

[0297] With this, an occupied region of the second roller pair **16** or the third roller pair **20** in a direction along the inter-axial distance $dk2$ can be reduced, and a device dimension can be reduced.

[0298] Next, the transmission substrate **52** and the reception substrate **55** are inclined with respect to the normal direction Da for the document surface S passing through between the transmission sensor chip **53** and the reception sensor chip **56**. The transmission surface $53a$ of the transmission sensor chip **53** extends along the surface of the transmission substrate **52**. The reception surface $56a$ of the reception sensor chip **56** extends along the surface of the reception substrate **55**.

[0299] Further, at least a part of the transmission substrate **52** and at least a part of the reception substrate **55** are within the range of the second roller pair **16** or the third roller pair **20** in the normal direction Da for the document surface S at the nipping position of the second roller pair **16** or the third roller pair **20**. With this, the device dimension in the normal direction Da can be reduced.

[0300] Note that, in the embodiment, the entire transmission substrate **52** and a part of the reception substrate **55** are on the inner side of the range **Ua2** in the normal direction **Da**. However, a part of the transmission substrate **52** and a part of the reception substrate **55** may be on the inner side of the range **Ua2** in the normal direction **Da**. Alternatively, a part of the transmission substrate **52** and the entire reception substrate **55** may be on the inner side of the range **Ua2** in the normal direction **Da**. Alternatively, the entire transmission substrate **52** and the entire reception substrate **55** may be on the inner side of the range **Ua2** in the normal direction **Da**.

[0301] Further, in the embodiment, the entire transmission sensor chip **53** and the entire reception sensor chip **56** are on the inner side of the range **Ua2** in the normal direction **Da**. However, a part of the transmission sensor chip **53** and a part of the reception sensor chip **56** may be on the inner side of the range **Ua2** in the normal direction **Da**. Alternatively, a part of the transmission sensor chip **53** and the entire reception sensor chip **56** may be on the inner side of the range **Ua2** in the normal direction **Da**. Alternatively, the entire transmission sensor chip **53** and a part of the reception sensor chip **56** may be on the inner side of the range **Ua2** in the normal direction **Da**.

[0302] Further, in the embodiment, a part of the reception substrate **55** and a part of the shielding member **91** are on the inner side of the range **Ua2** in the normal direction **Da**. With this, the device dimension in the normal direction **Da** can be reduced.

[0303] Note that, in the embodiment, a part of the reception substrate **55** and a part of the shielding member **91** are on the inner side of the range **Ua1** in the normal direction **Da**. However, the entire reception substrate **55** or the entire shielding member **91** may be on the inner side of the range **Ua1** in the normal direction **Da**.

[0304] Subsequently, with reference to FIG. 25, the configuration of the second roller pair **16** is further described.

[0305] In the embodiment, the two sets of the second roller pairs **16** are provided to extend along the X-axis direction, that is, the document width direction, and the two sets of the second roller pairs **16** are provided at symmetrical positions across a center position **CL** of the document **P** in the document width direction.

[0306] The reference symbol **17a** is a rotary shaft of the second lower roller **17**.

[0307] The reference symbol **18a1** is a rotary shaft located between the two second upper rollers **18**, and the reference symbol **18a2** is a rotary shaft located between the two second upper rollers **18**. The shaft diameter of the rotary shaft **18a1** is smaller than the shaft diameter of the rotary shaft **18a2**.

[0308] The rotary shaft **18a2** located in the +X direction is connected to a first universal joint **31A**. A second universal joint **31B** is further provided in the +X direction with respect to the first universal joint **31A**, and the first universal joint **31A** and the second universal joint **31B** are connected via the rotary shaft **18a3**.

[0309] The second upper roller **18** advances and retracts with respect to the second lower roller **17**, and a driving force is transmitted to the second upper roller **18**. Thus, the second upper roller **18** can be driven by the first universal joint **31A** and the second universal joint **31B** while advancing and retracting with respect to the second lower roller **17**.

[0310] The lower part of FIG. 25 illustrates a state in which the second upper roller **18** contacts with the second lower roller **17**. The upper part of FIG. 25 illustrates a state in which the second upper roller **18** is farthest from the second lower roller **17**. The reference symbol **M1** is a range within which the second upper roller **18** moves.

[0311] As illustrated, in the embodiment, the entire reception sensor chip **56** and the entire reception substrate **55** are on the inner side of the range **M1** in the normal direction **Da**. Thus, a device dimension in the normal direction **Da** can be reduced.

[0312] Note that, in the embodiment, the entire reception sensor chip **56** and the entire reception substrate **55** are on the inner side of the range **M1** in the normal direction **Da**. However, a part of the reception sensor chip **56** and a part of the reception substrate **55** may be on the inner side of the

range **M1** in the normal direction **Da**. Alternatively, the entire reception sensor chip **56** and a part of the reception substrate **55** may be on the inner side of the range **M1** in the normal direction **Da**.

[0313] Note that the relationship between the third roller pair **20**, and the reception sensor chip **56** and the reception substrate **55** is similar to the above-mentioned relationship between the second roller pair **16**, and the reception sensor chip **56** and the reception substrate **55**.

[0314] Note that, in the embodiment, the shaft diameter of the rotary shaft **17a** is 6.0 mm. Further, the shaft diameter of the rotary shaft **18a1** is 4.0 mm, the shaft diameter of the rotary shaft **18a2** is 5.0 mm, and the shaft diameter of the rotary shaft **18a3** is 4.5 mm. All of the shaft diameters thereof are greater than the thickness of the transmission sensor chip **53** or the reception sensor chip **56**.

[0315] Note that, as described above, the path length between the first roller pair **13** and the second roller pair **16**, in other words, the path length of the downstream feeding path **R1** is shorter than the path length between the second roller pair **16** and the third roller pair **20**, in other words, the path length of the reading transport path **R2**. With this, the path length between the first roller pair **13** and the second roller pair **16** can be reduced, and a device dimension can be reduced.

[0316] Further, in the embodiment, as described with reference to FIG. **13** to FIG. **15**, the transmission substrate **52** and the reception substrate **55** include the long sides **E2** and the short sides **E1**, and are provided so that the short sides **E1** extend along the transport direction **Fp**. With this, the first roller pair **13** and the second roller pair **16** are close to each other in the transport direction **Fp**, which can contribute to device size reduction.

[0317] Further, the lengths of the short sides **E1** of the transmission substrate **52** and the reception substrate **55** are shorter than a length of a glass plate **32e** in the transport direction **Fp**.

[0318] Here, configurations of the first reading unit **32** and the second reading unit **33** are described with reference to FIG. **26**. Note that the configurations of the first reading unit **32** and the second reading unit **33** are basically the same, and description is made below on the configuration of the first reading unit **32**.

[0319] The first reading unit **32** includes a lower housing **32f** and an upper housing **32g**, and includes a sensor substrate **32a** in the lower housing **32f**. The sensor substrate **32a** is provided with a line-like image sensor **32b** being an example of a reading sensor, and the image sensor **32b** receives reflection light from the document **P** via a lens **32c**. Note that the reference symbol **32d** is a light source that irradiates the document **P**.

[0320] The upper housing **32g** is provided with the glass plate **32e**. The glass plate **32e** transmits light reflected by the document **P** toward the lens **32c**. In addition, the glass plate **32e** includes a function of contacting with the document **P** and guiding the document **P** downstream.

[0321] Further, as described above, the lengths of the short sides **E1** of the transmission substrate **52** and the reception substrate **55** are shorter than the length of the glass plate **32e** in the transport direction **Fp**. Note that, in the embodiment, the lengths of the short sides **E1** of the transmission substrate **52** and the reception substrate **55** are 8.6 mm, the lengths of the long sides **E2** thereof are 26.8 mm, and the length of the glass plate **32e** in the transport direction **Fp** is 12.6 mm.

[0322] In this manner, the lengths of the short side **E1** of the transmission substrate **52** and the reception substrate **55** in the transport direction **Fp** can be reduced. As a result, the first roller pair **13** and the second roller pair **16** are closer to each other in the transport direction **Fp**, which can further contribute to device size reduction.

[0323] Note that, in the embodiment, the thickness of the glass plate **32e** is 1.1 mm, and is more than 0.578 mm, which is the thickness of the transmission sensor chip **53** or the reception sensor chip **56**.

[0324] Next, FIG. **27** illustrates the positional relationship between the transmission substrate **52** and the transmission sensor chip **53**, and the sensor substrate **32a** and the image sensor **32b** as the transmission substrate **52** and the transmission sensor chip **53** are viewed from upstream in the transport direction **Fp**. In FIG. **27**, the sensor substrate **32a** and the image sensor **32b** are indicated

by the two-dot chain line.

[0325] As illustrated, the transmission sensor chip **53** is located on the +Da side with respect to the image sensor **32b**, in other words, the second reading unit **33** side in the normal direction Da. With this, an amount by which the transmission sensor chip **53** protrudes to the -Da side, in other words, downward from the first reading unit **32** in the normal direction Da can be reduced, and a device dimension in the normal direction Da can be reduced.

[0326] Next, FIG. **28** illustrates the positional relationship between the transmission substrate **52** and the transmission sensor chip **53**, and the second reading unit **33** as the transmission substrate **52** and the transmission sensor chip **53** are viewed from upstream in the transport direction Fp. In FIG. **28**, the external shape of the second reading unit **33** and an image sensor **32B** provided to the second reading unit **33** are indicated by the two-dot chain line.

[0327] The second reading unit **33** advances and retracts with respect to the first reading unit **32**. The lower part of FIG. **28** illustrates the farthest state from the first reading unit **32**, and the upper part of FIG. **28** illustrates the most advanced state with respect to the first reading unit **32**. A range indicated with the reference symbol M2 is a region within which the second reading unit **33** is displaced.

[0328] As illustrated, at least a part of the reception sensor chip **56** is on the inner side of the range M2 in the normal direction Da. With this, the device dimension in the normal direction Da can be reduced.

[0329] Further, similarly, at least a part of the reception substrate **55** is on the inner side of the range M2 in the normal direction Da. With this, the device dimension in the normal direction Da can be reduced.

[0330] Note that, in the embodiment, the entire reception sensor chip **56** and the entire reception substrate **55** are on the inner side of the range M2 in the normal direction Da. However, a part of the reception sensor chip **56** and a part of the reception substrate **55** may be on the inner side of the range M2 in the normal direction Da. Alternatively, the entire reception sensor chip **56** and a part of the reception substrate **55** may be on the inner side of the range M2 in the normal direction Da.

[0331] Next, FIG. **29** illustrates occupied regions of the first roller pair **13**, the second roller pair **16**, the transport motor **47**, the transmission sensor chip **53**, the transmission substrate **52**, the reception sensor chip **56**, and the reception substrate **55** as viewed in the X-axis direction.

[0332] As illustrated, at least a part of the transmission substrate **52** and at least a part of the transmission sensor chip **53** are within the occupied region of the transport motor **47**. With this, an increase in the device size can be suppressed.

[0333] Note that, in the embodiment, a part of the transmission substrate **52** and the entire transmission sensor chip **53** are within the occupied region of the transport motor **47**. The entire transmission substrate **52** and the entire transmission sensor chip **53** may be within the occupied region of the transport motor **47**, or a part of the transmission substrate **52** and a part of the transmission sensor chip **53** may be within the occupied region of the transport motor **47**.

[0334] The present disclosure is not intended to be limited to the aforementioned exemplary embodiments, and many variations are possible within the scope of the present disclosure as described in the appended claims. It goes without saying that such variations also fall within the scope of the present disclosure.

[0335] For example, in the embodiment described above, the medium transport device **100** is applied to the scanner **1** being an example of the image reading apparatus, which is not limited thereto. the medium transport device **100** may be applied to a recording device that performs recording with respect to a medium or a post processing device that executes post processing such as staple processing, punching processing, or the like with respect to a medium.

Claims

1. A medium transport device comprising: a first roller pair configured to transport a medium in a transport direction; a second roller pair arranged downstream of the first roller pair in the transport direction; and an ultrasonic detection unit arranged between the first roller pair and the second roller pair in the transport direction, wherein the ultrasonic detection unit includes: a transmission sensor chip configured to emit an ultrasonic wave along a first axis toward a first surface of the medium being transported; a reception sensor chip being configured to receive an ultrasonic wave and being arranged at a position on the first axis so that the medium is sandwiched between the reception sensor chip and the transmission sensor chip; a transmission substrate being provided with the transmission sensor chip and including a substrate surface on which the transmission sensor chip is placed; a reception substrate being provided with the reception sensor chip and including a first substrate surface on which the reception sensor chip is placed; an amplifier being provided at the reception substrate and being configured to amplify a reception signal of an ultrasonic wave received by the reception sensor chip; and a shielding member being provided at the reception substrate and being configured to cover at least a part of the amplifier, and a height of the reception sensor chip with respect to the reception substrate is lower than a height of the shielding member with respect to the reception substrate.
2. A medium transport device according to claim 1, wherein the first axis is inclined with respect to a surface of the medium passing through between the transmission sensor chip and the reception sensor chip, a thickness of the transmission sensor chip is less than a thickness of the transmission substrate, a thickness of the reception sensor chip is less than a thickness of the reception substrate, and at least a part of the transmission sensor chip and at least a part of the reception sensor chip are present within a range of the second roller pair in a normal line direction with respect to a surface of the medium at a nipping position of the second roller pair.
3. A medium transport device according to claim 1, wherein the amplifier and the shielding member are provided to a substrate surface of the reception substrate, the substrate surface being opposite to the first substrate surface on which the reception sensor chip is placed.
4. A medium transport device according to claim 1, wherein the reception substrate is provided with a connector at a substrate surface opposite to the first substrate surface facing the transmission sensor chip, and the amplifier and the shielding member are arranged at a surface of the reception substrate, the surface being provided with the connector.
5. A medium transport device according to claim 1, wherein the amplifier is arranged at the first substrate surface of the reception substrate, the first substrate surface being a surface on which the reception sensor chip is placed, and the shielding member covers the reception sensor chip in addition to the amplifier.
6. A medium transport device according to claim 1, wherein at least a part of the reception substrate and at least a part of the shielding member are present within any one of: a range of the first roller pair in a normal line direction with respect to a surface of the medium at a nipping position of the first roller pair, and a range of the second roller pair in a normal line direction with respect to a surface of the medium at a nipping position of the second roller pair.
7. A medium transport device according to claim 1, wherein the reception sensor chip includes a piezoelectric element, the first axis is inclined with respect to a surface of the medium passing through between the transmission sensor chip and the reception sensor chip, and when the ultrasonic detection unit is calibrated, the reception sensor chip receives an ultrasonic wave while a bias voltage is applied to the reception sensor chip including the piezoelectric element, and the amplifier amplifies the reception signal.
8. A medium transport device according to claim 7, wherein the first axis is inclined with respect to a surface of the medium passing through between the transmission sensor chip and the reception sensor chip, and when the ultrasonic detection unit detects double feeding of media, the reception sensor chip receives an ultrasonic wave while a bias voltage is applied to the reception sensor chip,

and the amplifier amplifies the reception signal.

9. An image reading apparatus comprising: the medium transport device according to claim 1 and a reading unit being located downstream of the second roller pair in the transport direction and being configured to read an image of the medium.
